

Project 2

Building and Securing Personal Computer Systems

Scenario Options

For this assessment you must ground your decisions in a realistic context. Choose ONE of the following:

1. Horizon Analytics (Default) – A full scenario brief is supplied on the subject site.
2. Real-World Organisation – A small business, community group, or workplace you know personally. Provide a short description (≈100 words) covering staff size, core activities, and any special security concerns.
3. Original Fictional Scenario – Invent a plausible small office or start-up (≈100 words) including staff roles, growth expectations, and key security drivers.
4. Whichever option you choose, you will reference it throughout Parts 1-3. If you select option 2 or 3, include your scenario description in an appendix so markers can evaluate the appropriateness of your choices.

1.Part 1: PC Component Analysis

1.1. Activity: Specify a workstation suitable for staff use

Computer Use

E.g. gaming, multimedia creation, general, etc.

--

1.2. Visit BuildMyPC.net

- Navigate to [BuildMyPC.net](https://www.buildmypc.net).
- Go to the System Builder

1.3. Choose Your Components

- Select the following components:
 - CPU
 - GPU
 - RAM
 - Motherboard
 - Storage
 - Power Supply Unit (PSU)
 - Case

1.4. Analyse Each Component

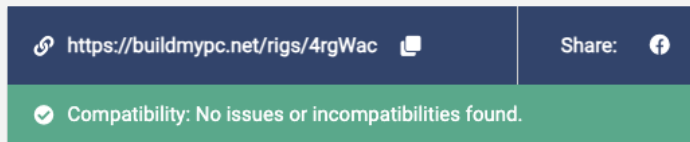
- For each component, answer the following:
 - What is its primary function? (E.g., The CPU processes instructions and executes tasks.)
 - What are the key specifications of the selected component? (E.g., CPU: 8 cores, 3.6 GHz clock speed.)
 - Why did you choose this component for your build? (E.g., The CPU was chosen for its multitasking capabilities.)

Choose your Components

Component	Function	Key Specifications	Justification
CPU			
GPU			
RAM			
Motherboard			
Storage			
PSU			
Case			

4. Copy Your Build Link

- Copy and paste the link to your build into your document



Link to build

2. Part 2: Cyber Security Measures

2.1. Activity: Securing a Personal Computer

Provide practical recommendations that an end-user can implement immediately. Organise your response under the headings below and justify each recommendation (why it is needed and which threat it mitigates).

Your task is to create a user-friendly information document by:

- Combining what you've learned in the following two sections:
 - **Part A: Securing the Computer:** Focus on practical, specific steps they can take with their device.
 - **Part B: Staying Safe Online:** Include general tips for using their computer safely in different environments (e.g., at home, on public Wi-Fi).
- Make your document clear and accessible for someone without a technical background.

2.1.1. Part A: Securing the Computer

- Research actions someone can take to secure their personal laptop. Focus on:
 - How to protect the laptop itself (e.g., preventing unauthorised access, securing data).
 - How to keep software and files safe from potential malware or hackers.
- As you research, consider:
 - Why are these actions necessary?
 - What specific threats or risks does each action address?

Hint: Think about tools or settings available on most computers, such as security features built into operating systems or additional software that enhances protection.

2.1.2. Part B: Staying Safe Online

- Think beyond the device itself. Research the habits or behaviours people should adopt to stay safe while using their computers. For example:
 - How can someone reduce risks when using public Wi-Fi?

Hint: Consider risks like phishing, identity theft, and data breaches. Look for tips that focus on human behaviour and decision-making.

3. Part 3 – Small Office Network Design & Security

3.1. Requirements Gathering

Summarise the business needs (number of staff/devices, services required, growth expectations).

3.2. Topology Diagram

Use the **Packet Tracer** screenshots to draw and label:

- ISP modem → firewall/router → switch → wired clients
- Wi-Fi access point(s) for wireless clients
- Optional network-attached storage (NAS) / printer

3.3. Device-Selection Table

<i>Device</i>	<i>Proposed Model</i>	<i>Estimated Cost (AUD)</i>	<i>Role in Network</i>	<i>Justification</i>
<i>Router / Firewall</i>				
<i>Switch</i>				
<i>Access Point</i>				
<i>NAS / File Server</i>				

3.4. IP-Addressing Scheme

Subnet	Purpose	CIDR / Mask	DHCP Range	Static Addresses	Reserved
--------	---------	-------------	------------	------------------	----------

LAN Workstations 192.168.x.0/24
WLAN Guest Visitors

3.5. Configuration Steps (Outline)

List the essential configuration steps you would perform on each device (e.g. change default admin credentials, create VLANs, enable WPA3, configure NAT & port-forwarding rules).

3.6. Network Security Recommendations

Provide at least **five** justified measures that extend beyond home-grade setups (e.g. deploying a UTM firewall, enabling 802.1X authentication, daily configuration backups).

3.7. Required & Recommended Resources

Meyers et al. (2019). *CompTIA IT Fundamentals+ All-in-One Exam Guide (FC0-U61)*. McGraw-Hill.

Calder (2020). *Cyber Security: Essential Principles to Secure Your Organisation*. ITGP.

Day (2017). *Security in the Digital World*. IT Governance Publishing.

Donaldson et al. (2018). *Understanding Security Issues*. De | G Press.

Hint: Referencing & Academic Integrity

- Cite all sources using **APA 7** style.
- Complete **Appendix A** to disclose any generative-AI assistance. Undeclared AI use will be treated as academic misconduct.

3.8. Software / Tools Used in Generating this Document

- Word Processor (MS Word, Google Docs)
- Coding/Dev Tools (VS Code, GitHub Desktop, etc.)
- diagrams.net
- Windows 11 PC with suitable CPU, GPU, RAM & storage

Appendix A – AI Disclosure Template

Tool(s) Used: e.g. ChatGPT (idea generation & outline structuring).

Extent of Use: e.g. Generated initial outline for Section 3.6; writer rewrote content for clarity.

Author Verification: All content reviewed & fact-checked by student.

Appendix B – AI Assessment Scale (excerpt)

- **Level 0:** No AI use...
- **Level 1:** AI-generated content with full re-draft...
- **Level 2:** AI-assisted idea generation & structuring (permitted for this assessment)...
- **Level 3:** Unedited AI output (prohibited)...

For the full scale, see subject-site Resources → Academic Integrity > AI Use Scale.